

Real Estate Price Prediction

Advanced Apex Project – Phase 1 Proposal

Team: The Outliers

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Course: Advanced Apex Project 1 (Trimester 1, 2025–26)

Submission: Phase 1 Proposal

Date: October 1, 2025

Project Title: Real Estate Price Prediction

1. Problem Statement

Accurate real estate price prediction is a major challenge for buyers, sellers, and investors. Traditional valuation methods rely on limited appraisal factors and often fail to capture the diverse influences such as lot size, number of rooms, construction year, location, and neighborhood characteristics. Inaccurate estimates may lead to poor investment decisions, buyer dissatisfaction, or overpricing in competitive markets. We need a data-driven approach to analyze historical housing data and build predictive models that can estimate house sale prices with greater reliability. This solution will also help identify which features most strongly affect property prices.

2. Business Goal

The primary goal is to develop a machine learning regression model that predicts property sale prices with strong accuracy and reliability. This will allow potential buyers and investors to make informed decisions, real estate agencies to improve their valuation practices, and stakeholders to understand the market drivers of property prices. By the end of the project, we aim to deliver a working predictive model along with clear insights and visualizations.

3. Data Source

We will use the **Ames Housing Dataset**, a widely used benchmark dataset for real estate price prediction.

Source Platform: Kaggle

Full Citation: Shashank Necrothapa. (n.d.). *Ames Housing Dataset* [Data set]. Kaggle. Retrieved October 1, 2025, from <https://www.kaggle.com/datasets/shashanknecrothapa/ames-housing-dataset>

Dataset Details:

- Instances: 2,930 observations
- Features: 79 attributes (numerical and categorical)
- Target: **SalePrice** (house sale price)
- Features describe lot size, rooms, year built, neighborhoods, etc.

4. Tools & Technologies

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Development Environment:** Jupyter Notebook
- **BI Tools:** Tableau or Power BI for final dashboard
- **Version Control:** GitHub

5. Project Workflow

The project will follow the standard data science lifecycle:

- **Data Acquisition:** Obtain the dataset from Kaggle (direct download method).
- **Preprocessing:** Handle missing values, clean inconsistencies, encode categorical variables.
- **Exploratory Data Analysis (EDA):** Use statistical summaries and visualizations to study patterns.
- **Feature Engineering:** Create new features (e.g., house age, price per sqft) and perform scaling/encoding.
- **Model Building:** Train regression models (Linear Regression, Ridge/Lasso, Random Forest, Gradient Boosting).
- **Model Evaluation:** Evaluate using RMSE, MAE, and R^2 .
- **Visualization & Reporting:** Create dashboards and storytelling slides highlighting findings.

6. Data Extraction

The dataset (Ames Housing) has been acquired using the **direct download method** from Kaggle. We manually downloaded the ZIP file from the Kaggle dataset page and extracted it into the project folder for further preprocessing.

Files: AmesHousing.csv **Location:** ./data/

This ensures that our dataset remains consistent with the official Kaggle version. A short inspection notebook will be prepared to confirm schema details and enable reproducibility in later phases.

7. Schema / Data Dictionary

A schema or data dictionary provides a structured overview of the dataset. For Phase 1, we prepared a concise dictionary with 20 key features that are most relevant for modeling house prices. The complete dictionary covering all 82 features of the Ames Housing dataset will be maintained separately in an Excel file (`data_dictionary.xlsx`) and updated as the project progresses.

Feature	Type	Description	PK
Id	Integer	Unique property identifier	Yes
MSSubClass	Categorical	Type of dwelling (e.g., 20 = 1-Story 1946+ All Styles, 30 = 1-Story 1945)	No
MSZoning	Categorical	General zoning classification (Residential, Commercial, etc.)	No
LotFrontage	Integer	Linear feet of street connected to property	No
LotArea	Integer	Lot size in square feet	No
Street	Categorical	Type of road access (Grvl, Pave)	No
LotShape	Categorical	General shape of property (Reg, IR1, IR2, IR3)	No
Neighborhood	Categorical	Physical location / neighborhood within Ames city	No
Condition1	Categorical	Proximity to main road or other conditions	No
BldgType	Categorical	Type of dwelling (1Fam, TwnhsE, etc.)	No
HouseStyle	Categorical	House style (1Story, 2Story, etc.)	No
OverallQual	Integer	Overall material and finish quality (scale 1–10)	No
OverallCond	Integer	Overall condition rating (scale 1–10)	No
YearBuilt	Integer	Original construction year	No
YearRemodAdd	Integer	Year of remodel/addition (if any)	No
GrLivArea	Integer	Above-grade (ground) living area square feet	No
FullBath	Integer	Full bathrooms above grade	No
BedroomAbvGr	Integer	Bedrooms above ground level	No
GarageCars	Integer	Garage capacity in number of cars	No
GarageArea	Integer	Garage size in square feet	No
SalePrice	Integer	Property sale price (Target variable)	No

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Phase: Phase 1 — Proposal

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